

ANJAN KUMAR THOTA

SENIOR DATA ENGINEER | PYTHON DEVELOPER

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PROFESSIONAL SUMMARY

Senior Data Engineer and Python Developer with 5+ years of experience designing, developing, and supporting scalable data pipelines, cloud-based data platforms, and backend applications. Experienced in Python, SQL, PySpark, Apache Spark, Kafka, Airflow, AWS, Linux, and Bash scripting. Strong background in ETL/ELT development, data integration, REST APIs, data warehousing, database design, SQL optimization, workflow automation, and data quality validation. Hands-on experience working with structured and unstructured data from databases, APIs, flat files, and cloud storage, building reliable data processing solutions, supporting analytics and reporting workloads, and automating recurring operational tasks. Skilled in Linux administration, shell scripting, cron scheduling, web scraping, database connectivity, and production support, with a proven ability to collaborate with business users, analysts, data scientists, and engineering teams to deliver secure, scalable, and maintainable data solutions.

TECHNICAL SKILLS

Programming & Scripting: Python, SQL, Bash, Shell Scripting

Data Engineering: ETL/ELT, Data Pipelines, Data Integration, Data Transformation, Data Wrangling, Data Warehousing, Data Quality

Big Data & Orchestration: PySpark, Apache Spark, Spark SQL, Apache Hadoop, HDFS, Hive, Kafka, Apache Airflow

Cloud Platforms: AWS (S3, Glue, EMR, Redshift, Athena, Lambda, EC2, IAM)

Databases & Modeling: PostgreSQL, MySQL, IBM Db2, Relational Databases, NoSQL, Database Design, Data Modeling, Stored Procedures, Views, Joins

Data Management: Data Quality, Data Integrity, Data Security, Data Governance, Data Architecture, Source-to-Target Validation

Backend & Integration: Django, Flask, FastAPI, REST APIs, Microservices, Web Scraping, Requests, BeautifulSoup, Data Import/Export

Analytics & Development: Pandas, NumPy, Scikit-learn, Jupyter Notebook, Tableau, Excel, Unit Testing

DevOps & Tools: Docker, Git, Jenkins, Terraform, JIRA, CI/CD, Linux, Unix, Cron, Vim, Nano

PROFESSIONAL EXPERIENCE

Senior Data Engineer | Healthfirst – New York, NY

Jul 2025 – Present

- Designed and implemented end-to-end batch and near-real-time data pipelines using PySpark, Spark SQL, and Kafka to ingest healthcare data from databases, APIs, files, and cloud storage into analytics-ready platforms.
- Built and maintained AWS-native data processing solutions using S3, Glue, EMR, Redshift, Athena, and Lambda, supporting scalable data lake, data warehouse, reporting, and downstream analytics requirements.
- Developed reusable ingestion and transformation frameworks for structured and unstructured datasets, incorporating schema validation, standardization, deduplication, exception handling, and configurable

business rules.

- Created and supported Apache Airflow DAGs with task dependencies, scheduling, retries, backfills, failure notifications, and operational controls to ensure reliable and timely delivery of critical datasets.
- Optimized Spark workloads and complex SQL queries using partitioning, caching, broadcast joins, efficient file sizing, predicate pushdown, and query-plan analysis to improve processing efficiency and resource utilization.
- Implemented automated data-quality controls covering completeness, accuracy, uniqueness, referential integrity, source-to-target reconciliation, and invalid-record handling across production data pipelines.
- Established logging, monitoring, operational dashboards, and alerting processes that enabled engineering teams to identify failed jobs, investigate data issues, and perform root-cause analysis more efficiently.
- Prepared curated datasets and feature-engineering workflows in partnership with analytics and data science teams, enabling machine learning, regulatory reporting, business intelligence, and operational decision-making use cases.
- Applied IAM roles, encryption, least-privilege access, and role-based permissions to protect sensitive information while supporting organizational data governance, compliance, and audit requirements.
- Containerized pipeline components with Docker and contributed to Git-based development, CI/CD deployments, peer code reviews, production support, technical documentation, and continuous improvement of engineering standards.

Environment: Python, PySpark, Spark SQL, Kafka, Airflow, AWS S3, Glue, EMR, Lambda, Redshift, Athena, Docker, Git, Tableau

Python Developer | Ryder System, Inc. – Fort Worth, TX

Jul 2024 – Jun 2025

- Gathered business requirements and converted functional needs into technical designs, backend workflows, data models, and maintainable Python solutions aligned with application and reporting objectives.
- Designed, developed, and supported backend applications and internal services using Python, Django, and Flask, following modular development practices to improve maintainability, testability, and reuse.
- Built RESTful APIs and microservices for communication among web applications, enterprise platforms, reporting systems, and third-party services, including input validation, structured error handling, and secure data exchange.
- Developed Python and shell-based extraction utilities to collect data from Linux servers, relational databases, APIs, and flat files, then transform and load the results into Hadoop HDFS and analytics environments.
- Automated recurring reporting and data-processing workflows using Python, cron jobs, Pandas, NumPy, SQL, and Spark SQL, reducing dependence on manual data preparation and improving consistency across scheduled outputs.
- Developed and maintained Bash shell scripts to automate Linux-based operational tasks, file transfers, scheduled jobs, log monitoring, backup processes, and data-processing workflows.
- Created and optimized SQL queries, joins, stored procedures, and database interactions for CRUD operations, reporting, and application workflows while addressing performance bottlenecks in backend processes.
- Implemented logging, exception handling, retry mechanisms, validation rules, and job-status reporting to strengthen operational reliability and simplify troubleshooting of automated processes and data integrations.

- Integrated external APIs and enterprise systems while implementing authentication, authorization, session controls, and secure handling of application data in accordance with internal development standards.
- Developed Tableau dashboards for operational KPIs and collaborated in unit testing, code reviews, deployments, production incident analysis, defect resolution, Agile ceremonies, and technical knowledge transfer.

Environment: Python, Django, Flask, REST APIs, SQL, Hadoop, HDFS, Spark, Pandas, NumPy, Tableau, Linux, Git, JIRA

Data Analyst / Data Engineer | IDS (Information Data Systems) – Hyderabad, India

Sep 2018 – Jun 2022

- Analyzed structured and unstructured datasets to identify trends, patterns, data-quality concerns, and business insights that supported operational reporting, planning, and stakeholder decision-making.
- Created Python-based extraction and transformation processes for XML, CSV, and relational data sources, converting raw information into standardized datasets suitable for reporting and analytical use.
- Performed data cleaning and preprocessing with Pandas and NumPy, including missing-value treatment, outlier review, type conversion, deduplication, normalization, and validation of business rules.
- Designed and maintained MySQL schemas, tables, stored procedures, indexes, and SQL queries to support reliable data storage, recurring reports, ad hoc analysis, and application-driven data requirements.
- Developed repeatable data-preparation, reconciliation, and validation workflows that improved consistency between source systems, reporting tables, analytical datasets, and stakeholder-facing outputs.
- Conducted exploratory data analysis and supported existing predictive models by preparing features, reviewing model inputs and outputs, and applying techniques such as logistic regression and decision trees.
- Partnered with business users to define KPIs, reporting logic, and acceptance criteria, then developed SQL-based reports, Excel outputs, and Tableau dashboards that communicated results clearly.
- Documented data models, transformation rules, stored procedures, analytical methods, and dashboard definitions while supporting issue investigation, knowledge transfer, and ongoing enhancement requests.

Environment: Python, SQL, MySQL, Pandas, NumPy, Scikit-learn, Tableau, Excel, Linux

EDUCATION

Master's in Information Technology Management

Webster University – St. Louis, MO

Aug 2022 – May 2024

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- Introduction to Data Engineering – IBM via Coursera
- Python Project for Data Engineering – IBM via Coursera
- Databases and SQL for Data Science with Python – IBM via Coursera
- Python for Data Science, AI & Development – IBM via Coursera
- Introduction to Relational Databases (RDBMS) – IBM via Coursera
- Hands-on Introduction to Linux Commands and Shell Scripting – IBM via Coursera